We create chemistry

# Safety data sheet



## 1. Identification

### Product identifier

## Bantux 500

### Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: crop protection product, herbicide

### Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Crop Protection

Telephone: +49 621 60-27777
E-mail address: Produktinformation-Pflanzenschutz@basf.com

### Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## 2. Hazards Identification

### Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 4 (oral)
Carc. 2
Aquatic Acute 1
Aquatic Chronic 1

For the classifications not written out in full in this section the full text can be found in section 16.

## Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Warning

Hazard Statement:

| | |
|---|---|
| H302 | Harmful if swallowed. |
| H351 | Suspected of causing cancer. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read carefully and follow all instructions. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P201 | Obtain special instructions before use. |
| P202 | Do not handle until all safety precautions have been read and understood. |
| P270 | Do not eat, drink or smoke when using this product. |
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P264 | Wash contaminated body parts thoroughly after handling. |

Precautionary Statements (Response):

| | |
|---|---|
| P308 + P313 | IF exposed or concerned: Get medical attention. |
| P301 + P312 | IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell. |
| P330 | Rinse mouth. |
| P391 | Collect spillage. |

Precautionary Statements (Storage):

| | |
|---|---|
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Labeling of special preparations (GHS):
May produce an allergic reaction. Contains: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide, 2-Methyl-2H-isothiazol-3-one, 1,2-Benzisothiazol-3(2H)-one

According to UN GHS criteria

Hazard determining component(s) for labelling: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide

### Other hazards

According to UN GHS criteria

See section 12 - Results of PBT and vPvB assessment.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

## 3. Composition/Information on Ingredients

### Substances

Not applicable

### Mixtures

Chemical nature

crop protection product, herbicide, suspension concentrate (SC)

Hazardous ingredients (GHS)
According to UN GHS criteria

metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide

| | |
|---|---|
| Content (W/W): 43,55 % | Acute Tox. 5 (oral) |
| CAS Number: 67129-08-2 | Skin Sens. 1B |
| EC-Number: 266-583-0 | Carc. 2 |
| | Aquatic Acute 1 |
| | Aquatic Chronic 1 |
| | M-factor acute: 100 |
| | M-factor chronic: 100 |
| | H303, H317, H351, H400, H410 |

Benzenesulfonic acid, hydroxy-, polymer with formaldehyde, phenol and urea, sodium salt

| | |
|---|---|
| Content (W/W): < 5 %<br>CAS Number: 102980-04-1 | Eye Dam./Irrit. 2A<br>Aquatic Acute 3<br>Aquatic Chronic 3<br>H319, H402, H412 |

1,2-benzisothiazol-3(2H)-one

| | |
|---|---|
| Content (W/W): < 0,005 %<br>CAS Number: 2634-33-5<br>EC-Number: 220-120-9<br>INDEX-Number: 613-088-00-6 | Acute Tox. 2 (Inhalation - dust)<br>Acute Tox. 4 (oral)<br>Skin Irrit. 2<br>Eye Dam. 1<br>Skin Sens. 1A<br>Aquatic Acute 1<br>Aquatic Chronic 1<br>M-factor acute: 1<br>M-factor chronic: 1<br>H318, H315, H330, H302, H317, H400, H410<br><br><u>Specific concentration limit:</u><br>Skin Sens. 1A: >= 0,036 % |

2-Methyl-2H-isothiazol-3-one

| | |
|---|---|
| Content (W/W): < 0,005 %<br>CAS Number: 2682-20-4<br>EC-Number: 220-239-6<br>INDEX-Number: 613-326-00-9 | Acute Tox. 2 (Inhalation - dust)<br>Acute Tox. 3 (oral)<br>Acute Tox. 3 (dermal)<br>Skin Corr. 1B<br>Eye Dam. 1<br>Skin Sens. 1A<br>Aquatic Acute 1<br>Aquatic Chronic 1<br>M-factor acute: 10<br>M-factor chronic: 1<br>H330, H317, H314, H301 + H311, H400, H410<br>EUH071<br><br><u>Specific concentration limit:</u><br>Skin Sens. 1A: >= 0,0015 % |

Propane-1,2-diol

Content (W/W): < 10 %
CAS Number: 57-55-6
EC-Number: 200-338-0

For the classifications not written out in full in this section the full text can be found in section 16.

## 4. First-Aid Measures

### Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

### Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

### Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

### Extinguishing media

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

### Special hazards arising from the substance or mixture

Carbon monoxide, hydrogen chloride, Carbon dioxide, nitrogen oxides, halogenated compounds, sulfur oxides, silicon oxides
The substances/groups of substances mentioned can be released in case of fire.

### Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## 6. Accidental Release Measures

### Personal precautions, protective equipment and emergency procedures

Do not breathe vapour/spray. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

### Environmental precautions

Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

### Methods and material for containment and cleaning up

For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

## 7. Handling and Storage

### Precautions for safe handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
No special precautions necessary. The substance/product is non-combustible. Product is not explosive.

### Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 60 Months

Protect from temperatures below: -5 °C
Changes in the properties of the product may occur if substance/product is stored below indicated temperature for extended periods of time.
Protect from temperatures above: 40 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

### Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## 8. Exposure Controls/Personal Protection

### Control parameters

Components with occupational exposure limits

57-55-6: Propane-1,2-diol
67129-08-2: 2-Chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)acetamide
TWA value 1,13 mg/m3 (BASF recomm. occupational exposure limit)

### Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for higher concentrations or long-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic and alkaline compounds (e.g. EN 14387 Type ABEK).

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## 9. Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| State of matter: | liquid | |
| Form: | liquid | |
| Colour: | white to pink | |
| Odour: | faint odour, fruity | |
| Odour threshold: | Not determined due to potential health hazard by inhalation. | |
| crystallization temperature: | approx. -8 °C | (measured) |
| boiling temperature: | approx. 100 °C (1.013 hPa) | (measured) |
| Flammability: | not highly flammable | (Directive 92/69/EEC, A.12) |
| Lower explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. | |

| | | |
|---|---|---|
| Upper explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. | |
| Flash point: | No flash point - Measurement made up to the boiling point. | (Directive 92/69/EEC, A.9) |
| Auto-ignition temperature: | 552 °C | (Directive 92/69/EEC, A.15) |
| Thermal decomposition: | No decomposition if stored and handled as prescribed/indicated. | |
| pH value: | approx. 5 - 8<br>(approx. 20 °C)<br>(measured with the undiluted substance) | (pH Meter) |
| Viscosity, dynamic: | approx. 82 mPa.s<br>(20 °C, 100 1/s) | (OECD Guideline 114) |
| Solubility in water: | dispersible | |
| Partitioning coefficient n-octanol/water (log Kow): | The statements are based on the properties of the individual components. | |

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*
*Partitioning coefficient n-octanol/water (log Kow): 2,13*
*(22 °C)*

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| | | |
|---|---|---|
| Vapour pressure: | approx. 23 hPa<br>(20 °C)<br>Information applies to the solvent. | |
| Density: | approx. 1,15 g/cm3<br>(20 °C)<br>approx. 1,14 g/cm3<br>(55 °C) | (Directive 92/69/EEC, A.3) |
| Relative vapour density (air): | not applicable | |

## 9.2. Other information

**Information with regard to physical hazard classes**

Explosives

| | | |
|---|---|---|
| Explosion hazard: | not explosive | (Directive 92/69/EEC, A.14) |

Oxidizing properties

| | | |
|---|---|---|
| Fire promoting properties: | not fire-propagating | (Directive 2004/73/EC, A.21) |

**Other safety characteristics**

| | |
|---|---|
| Other Information: | If necessary, information on other physical and chemical parameters is indicated in this section. |
| Evaporation rate: | not applicable |

## 10. Stability and Reactivity

### Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

### Chemical stability

The product is stable if stored and handled as prescribed/indicated.

### Possibility of hazardous reactions

No hazardous reactions if stored and handled as prescribed/indicated.

### Conditions to avoid

See SDS section 7 - Handling and storage.

### Incompatible materials

Substances to avoid:
strong oxidizing agents, strong bases, strong acids

### Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of moderate toxicity after single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation.

Experimental/calculated data:
LD50 rat (oral): > 500 - < 2.000 mg/kg (OECD Guideline 423)

LC50 rat (by inhalation): > 6,2 mg/l 4 h
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

LD50 rat (dermal): > 4.000 mg/kg
No mortality was observed. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Irritation

Assessment of irritating effects:
Not irritating to eyes and skin.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation rabbit: non-irritant
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
Guinea pig maximization test guinea pig: Non-sensitizing.

Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*
*Assessment of carcinogenicity:*
*Indication of possible carcinogenic effect in animal tests.*
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Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

Specific target organ toxicity (single exposure)

Assessment of STOT single:

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*
*Assessment of repeated dose toxicity:*
*Repeated exposure to large quantities may affect certain organs. Based on available data, the classification criteria are not met.*
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Aspiration hazard

not applicable

Other relevant toxicity information

Misuse can be harmful to health.

## 12. Ecological Information

### Toxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.

Aquatic plants:
ErC50 (72 h) 0,075 mg/l, Ankistrodesmus bibraianus

EC10 (72 h) 0,005 mg/l, Ankistrodesmus bibraianus

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*
*Toxicity to fish:*
*LC50 (96 h) 8,5 mg/l, Oncorhynchus mykiss*
---

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*
*Aquatic invertebrates:*
*EC50 (48 h) 33,7 mg/l, Daphnia magna*
---

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*

*Chronic toxicity to fish:*
*No observed effect concentration (28 d) 2,15 mg/l, Oncorhynchus mykiss*
---

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*
*Chronic toxicity to aquatic invertebrates:*
*No observed effect concentration (21 d) 2,56 mg/l, Daphnia magna*
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## Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*
*Assessment biodegradation and elimination (H2O):*
*Not readily biodegradable (by OECD criteria).*
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## Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*
*Assessment bioaccumulation potential:*
*No significant accumulation in organisms is expected as a result of the distribution coefficient of n-octanol/water (log Pow).*
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## Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: metazachlor (ISO); 2-chloro-N-(2,6-dimethylphenyl)-N-(1H-pyrazol-1-ylmethyl)-acetamide*
*Assessment transport between environmental compartments:*
*Adsorption in soil: Following exposure to soil, the product trickles away and can - dependant on degradation - be transported to deeper soil areas with larger water loads.*
---

## Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

### Other adverse effects

The product does not contain substances that are listed in the Montreal Protocol on substances that deplete the ozone layer.

### Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

### Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## 14. Transport Information

**Land transport**

ADR

| | |
|---|---|
| UN number or ID number: | UN3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (METAZACHLOR) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

RID

| | |
|---|---|
| UN number or ID number: | UN3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (METAZACHLOR) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |

| | |
|---|---|
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| UN number or ID number: | UN3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (METAZACHLOR) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

Transport in inland waterway vessel

Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (METAZACHLOR) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | EmS: F-A; S-F |

**Air transport**

IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (METAZACHLOR) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

## Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

**Further information**
Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2) and also the Special Provision 375 in Appendix B which is regulated in China "Regulations Concerning Road Transportation of Dangerous Goods Part 3: Index of dangerous goods name and transportation requirements" (JT/T 617.3)

## 15. Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

To avoid risks to man and the environment, comply with the instructions for use.

## 16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Acute Tox. | Acute toxicity |
| Carc. | Carcinogenicity |
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| Skin Sens. | Skin sensitization |
| Eye Dam./Irrit. | Serious eye damage/eye irritation |
| Skin Irrit. | Skin irritation |
| Eye Dam. | Serious eye damage |
| Skin Corr. | Skin corrosion |
| H303 | May be harmful if swallowed. |
| H317 | May cause an allergic skin reaction. |
| H351 | Suspected of causing cancer. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |
| H319 | Causes serious eye irritation. |
| H402 | Harmful to aquatic life. |
| H412 | Harmful to aquatic life with long lasting effects. |
| H318 | Causes serious eye damage. |
| H315 | Causes skin irritation. |
| H330 | Fatal if inhaled. |
| H302 | Harmful if swallowed. |
| H314 | Causes severe skin burns and eye damage. |
| H301 + H311 | Toxic if swallowed or in contact with skin. |
| EUH071 | Corrosive to the respiratory tract. |

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the

responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.